

RAJAT SINGH

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Professional Summary

Full-stack fresher who ships real, working products rather than tutorials — React frontends, Node.js/FastAPI backends, and AI-powered features built and shipped end-to-end. Comfortable using AI tools and LLM pipelines to build, debug, and validate faster, and equally comfortable reviewing that AI-generated output carefully before it ships. Security-first instincts (zero-knowledge encryption, token-based auth, input validation) and a strong sense of UI/UX from building products real users actually open. Curious, fast to learn, and not limited by "that's not my stack."

Key Projects

Huddle — Real-Time Chat Platform — React, Node.js, Express, Socket.IO, PostgreSQL

[GitHub](#) | [Demo](#)

- Built a full-stack real-time chat platform (React 19 + Vite frontend, Node.js/Express + Socket.IO backend, PostgreSQL) with time-boxed rooms, REST APIs, and a clean, minimal UI for room creation, joining, and live messaging.
- Implemented Google ID token authentication with server-side verification and an admin dashboard for user/room oversight; containerized PostgreSQL with Docker Compose for reproducible local development.

Lockyt — Password Manager — React, Firebase, Chrome Extension (MV3)

[GitHub](#) | [Demo](#)

- Designed a security-first, zero-knowledge password manager (React, Firebase) using AES-256-GCM client-side encryption and PBKDF2 key derivation (600,000 iterations), ensuring 0 bytes of plaintext ever reach the server across 5 vault item types.
- Shipped a companion Chrome Extension (MV3) with sub-second autofill and a 4-module security-analysis engine (breach detection, password-strength scoring, reuse detection, expiry alerts) — secure-by-design thinking applied end-to-end.

FrameForge (Manim Animation Generator) — React, FastAPI, LLM Agents, Supabase

[GitHub](#) | [Demo](#)

- Built a full-stack AI-powered app (React frontend, FastAPI backend) that turns natural-language prompts into rendered animations, using an LLM pipeline (Groq Llama 3.3 70B) with an intent-verification stage and an 8-pass validation/auto-fixer harness that reviews and repairs AI-generated code before it runs.
- Reduced generation failures by 90% and cut average latency from 60s to under 2s for 9 high-frequency prompt types via a template cache with LLM intent verification — hands-on experience building and carefully reviewing AI-agent-generated output, not just prompting it.

Document Grounded QA System — Python, LangChain, FAISS, Streamlit

[GitHub](#) | [Demo](#)

- Built an AI-powered RAG Q&A tool processing 130+ page documents with 95% accuracy and <3s query latency, using LangChain and FAISS with optimized chunking (1500/300 tokens) for 40% improved retrieval precision — a side project exploring how far agentic retrieval tools can go.
- Implemented output-verification checks that cut hallucinations by 60% and raised source-attribution accuracy to 98%, applying the same "review AI output before you trust it" discipline to a personal project.

Experience

AI/Cloud Intern — IBM SkillsBuild

Jul 2025 – Aug 2025

Built conversational AI systems using IBM Watson and Watson Assistant with context management, intent classification, and entity recognition, and integrated AI models with enterprise workflows and cloud infrastructure on IBM Cloud.

Education

University of Delhi — M.Sc. Computer Science

2026 – Present

University of Delhi — BSc (Hons) Computer Science, 8.5 GPA

2022 – 2026

Technical Skills

Languages: JavaScript, Python, C++, SQL

Frontend: React.js, Vite, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, Socket.IO

Auth & Security: JWT, AES-256-GCM Encryption, PBKDF2, Input Validation, Zero-Knowledge Design

Databases & Cloud: PostgreSQL, MongoDB, Supabase, Firebase, AWS (EC2, S3)

AI Tools & Agents: LangChain, LLM Tool-Calling, Prompt Evaluation & Validation, FAISS

DevOps & Tools: Docker, Git, Linux, Postman

Certificates

- Agentic AI Certified Foundations Associate, Oracle — 2026 [Credentials](#)
- Mastering Generative AI and ChatGPT, GeeksforGeeks — 2025 [Credentials](#)

Achievements

- Solved 350+ Data Structures and Algorithms problems on [Leetcode](#) and 200+ on [GeeksforGeeks](#)